

A Comparative Study of Perceptron-Based Models for Diabetes Prediction

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Abstract

The work presents a comparative analysis of multiple Perceptron-based models applied to the prediction of diabetes using the PIMA Indian diabetes dataset. The paper looks into the performances of the Baseline Perceptron, Stochastic Perceptron, Weighted Perceptron, Perceptron with Synthetic Minority Oversampling Technique (SMOTE), along with the Multilayer Perceptron (MLP). Each model was assessed based on performance indicators such as train and test accuracies, sensitivity, specificity, train and test losses, false negative, false positive, and false discovery rates aimed at varying the learning rates, penalties, and activation functions (in the case of MLP). The results were noticeably different in each iteration of variances of tuning parameters like learning rates, penalties, and activation functions for various experiments tested on an imbalanced dataset used in this study. The research showcases the strengths and weaknesses of deploying a perceptron-based model for detecting early diabetes.

1. Background

Frank Rosenblatt invented the perceptron in the 1950s as part of the key effort toward developing a computer model that truly mimicked human neurons. That was to be the origin of artificial neural networks since it came from the greater computational effort to understand cognition. The necessity for constructing a machine learning algorithm that adapts itself using examples was answered by this simple yet attractive approach, making it feasible to achieve other sorts of complex artificial intelligence (AI) [1,2].

The perceptron gained significance in machine learning as it introduced the notion of trainable weights, (i.e. based on the input data, the model can improve and adapt its performance). As a result, learning algorithms and complex neural network architectures were developed [3].

Figure 1 illustrates the components of a perceptron where Weight updates are made according to a simple adjustment rule called the perceptron learning algorithm. This makes it possible for this system to amend errors arising from differences between predictions made

against observed outcomes based on training data. Despite earlier evidence showing that single-layer perceptrons cannot handle linearly inseparable patterns; it has developed into multiple-layer perceptrons (MLPs). By being made up of many interconnected perceptrons at each level, MLPs can approximate highly intricate nonlinear functions thus significantly expanding the set of problems they can solve [2,4].

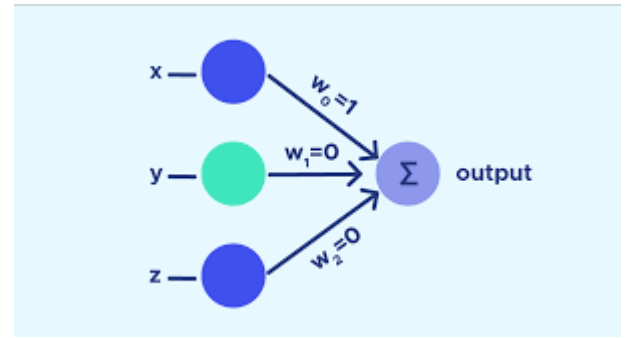


Figure 1. **Basic Perceptron Structure** showing inputs, weights, summation, activation function, and output.

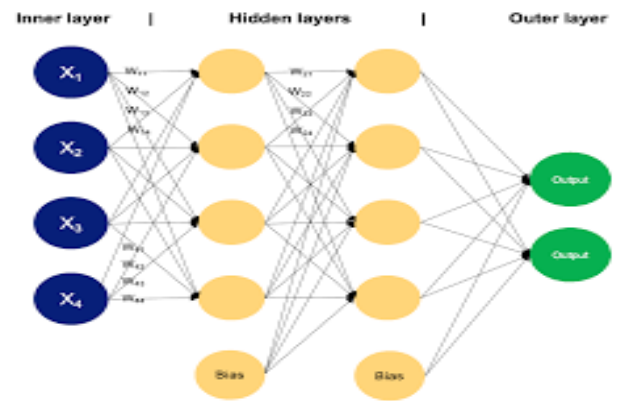


Figure 2. **Structure of Multilayer Perceptron** showing input layer, hidden layers, and output layer.

In the field of medicine, one can predict if a patient will get diabetes by utilizing perceptrons or multi-layer perceptrons in finding out outcomes. A multilayer perceptron includes an input layer, several hidden layers, and an output layer as given in Figure 2, where neurons of each layer are connected with the ones in the forward and backward layers. It learns to model complicated

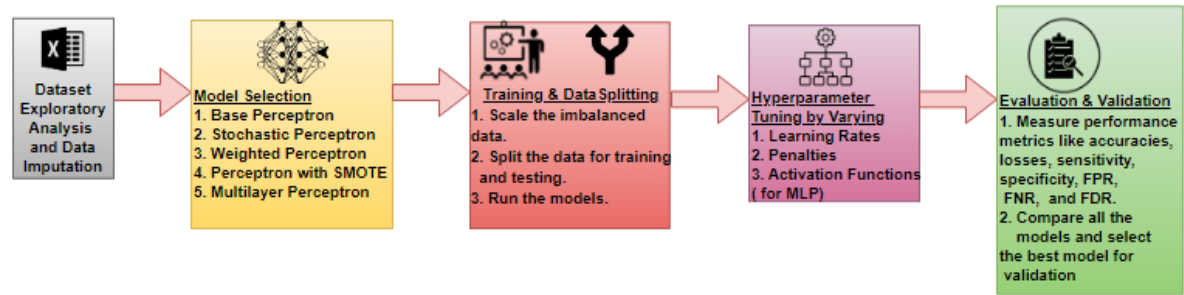


Figure 3. **Workflow Diagram** contains the steps followed to evaluate different types of perceptron model in diabetes prediction with metrics.

relationships by adjusting the weights between different neurons in such a way that turns out to be very effective in supervising how it can learn classification and regression. Due to the ability to learn from examples and generalize to new data, Perceptron became one of the valuable tools in pattern recognition and decision-making across various fields [3, 5].

This single classifier has evolved from the basic classifier we have seen above to being a component of complex neural networks. However, we have always had such fundamental principles with which we have been pushing forward with technology – some rather simple, key notions appear to be fundamental to the sort of revolutionary changes for computational intelligence [5].

2. Method Description

2.1 Dataset Description

The PIMA Indian Diabetes Dataset has been used for the aim of diabetes prediction using Perceptron. The dataset contained features such as Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, Body Mass Index, Diabetes Pedigree Function (function for predicting the likelihood of diabetes based on family history) [6], Age, and outcomes (0 for non-diabetic and 1 for diabetic). The dataset consisted of 768 samples with eight features for describing medical details and one for outcome. Certain data points, such as blood pressure, skin thickness, and insulin, have missing values.

2.2 Brief Overview

The principal objective of these studies is to investigate diverse perceptron-based models, both linear and non-linear, and evaluate their capacity to manage class imbalance, accuracy, and error rates. The research also assesses the influence of hyperparameter tuning (learning rates, penalties, and activation functions) [7], on model performance, specifically regarding critical metrics like as accuracy, loss, sensitivity, specificity, and error rates like false positive, negative, and discovery rates [8].

In this presented study, the dataset used underwent extensive exploratory dataset analysis to find missing or

anomalous values, such as zero, in fields like the insulin level since this is not a biologically valid reading. The missing value detected was imputed by replacing them with the median of respective columns and hence maintained the integrity and consistency of the dataset [9]. This was one very important step toward preparing data for subsequent modeling experiments.

This was further put through five experiments with varied models based on the perceptron as given in Figure 3, including:

i) Baseline Perceptron: This served as a foundational model in the classification task using a simple linear decision boundary without advanced optimization techniques [11].

ii) Stochastic Perceptron: Perceptron model in which weights get updated after each single data point [12].

iii) Weighted Perceptron: This model modifies the weights to account for class imbalance, ensuring that the minority class, which relates to diabetic patients is considered. [10, 11]

iv) Perceptron with Synthetic Minority Oversampling Technique (SMOTE): SMOTE was applied before training for the creation of artificial examples of the minority class. This method will make the model more robust against imbalanced data [13].

v) Multilayer Perceptron: A sophisticated neural network architecture for capturing nonlinear relations in data. MLP uses multiple layers of neurons and is more sophisticated in capturing intricate patterns in a dataset [5].

These models were tuned by varying:

a. Learning rates: This is the rate at which an algorithm learns or updates the values of a parameter estimate(weights). In the experiments, the learning rates of [0.001, 0.01, 0.1] were varied [7].

b. Penalties: These are regularization techniques that are used in algorithms and machine learning modeling with the view to preventing overfitting by adding weights to the loss function [14]. The following are the varied penalty types:

i) L1 Regularization: It is also called Lasso. It is used in adding the absolute values of the weights to the loss functions, which would help the model to be more capable by making irrelevant weights reach 0 [14].

ii) L2 Regularization: It is known as Ridge. It helps in adding the squared values of the weights to the loss functions without making unnecessary weights approach 0, but discouraging the large weights [14].

iii) Elastic Net: The technique, which is a combination of both L1 and L2, maintains a proper balance between scatteredness without allowing any large weights [14].

c. Activation Functions: Non-linearity will be provided to the neural network by the activation function, which helps it to learn and be able to represent complex patterns in the data. Activation functions are tested only on MLP since the regularization is handled by default with the use of some penalties like dropout, and L2 on all the weights of the network rather than particularly applying on weights that are interconnected between input and output layers which is the case for linear perceptrons. In the linear models such as baseline, stochastic, weighted, and smote, the default activation function was the sign function. In the MLP, several activating functions were tried in order to see how they would impact the performance of the model [7, 15].

i) Identity: This is a very simple linear function passing the input as it is without any transformation. It is used mainly where no non-linearity is required [7].

ii) Logistic: It is one of the most used activation functions, which squashes the input in the range between 0 and 1; hence, it is good for binary classification [7].

iii) Tanh - hyperbolic tangent: This has a similar shape to the sigmoid function. However, it has outputs in the range between -1 and +1. Therefore, this function centers the data better [7].

iv) ReLU: stands for Rectified Linear Unit. It represents the most-used activation function that offers an output of 0 for the region of negative inputs and returns the exact value in case of positive inputs. Due to the processing speed, it is computationally efficient and helps to reduce the problem of a vanishing gradient [7].

The models with the variation of the mentioned parameters were validated with their results from confusion matrices which contained True Positives, True Negatives, False Positives, and False negatives [16], for each experiment with the following key metrics:

I) Accuracies: A Measure of the total number of real diabetic patients out of the total number of patients. Accuracies were calculated with the `accuracy_score()` function in Python for test and train data samples [17].

II) Losses: A measure of the difference between the predicted values and the actual labels which is calculated with the `log_loss()` function in Python for test and training data samples [17].

III) Sensitivity: A Measure of the total number of actual diabetes patients that were correctly identified by the model [16].

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

IV) Specificity: A Measure of the total number of actual nondiabetic patients that were correctly identified by the model [16].

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

V) False Positive Rate (FPR): A Measure of the non-diabetic patients, who were incorrectly marked diabetic by the model [16].

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

VI) False Negative Rate (FNR): A Measure of the diabetic patients, who were incorrectly marked as non-diabetic [16].

$$\text{FNR} = \frac{\text{False Negatives}}{\text{False Negatives} + \text{True Positives}}$$

VII) False Discovery Rate (FDR): A Measure of false positives among the total positive predictions [16].

$$\text{FDR} = \frac{\text{False Positives}}{\text{True Positives} + \text{False Positives}}$$

These metrics helped determine the competencies of the experimented Perceptron models.

3. Experimental Analysis

3.1 Overall Perspective

The experiments made use of sci-kit learn(sk-learn) to train and assess the performances of models. The baseline, stochastic, and weighted Perceptrons were implemented using the Perceptron class from sk-learn kit, which can tune the model's hyperparameters, such as learning rates and penalties. The functions that were used for analyzing such as `accuracy_score`, `log_loss`, and `confusion_matrix` were components of the sk-learn toolkit [18]. The hyperparameters were tuned on the experiments by splitting the dataset into 80 percent for training and 20 percent for training as given in Table 1. The learning rates and penalties varied across all the experiments, while the activation functions varied for Multilayer Perceptron.

The set of experiments in this paper is driven to evaluate the performance of various perceptron-based machine learning models proposed toward class imbalance for the diagnosis of diabetes in patients using an imbalanced class dataset. Diabetes prediction is one of the important applications in health care, and the correct identification of diabetic patients is very important. However, minimizing false positives or non-diabetic patients labeled as diabetes is equally important.

Experiment Name	Hyperparameters Varied
Baseline Perceptron	Learning Rates: 0.001, 0.01, 0.1 Penalties: L1, L2, Elastic Net
Stochastic Perceptron	Learning Rates: 0.001, 0.01, 0.1 Penalties: L1, L2, Elastic Net
Weighted Perceptron	Learning Rates: 0.001, 0.01, 0.1 Penalties: L1, L2, Elastic Net
Perceptron With SMOTE	Learning Rates: 0.001, 0.01, 0.1 Penalties: L1, L2, Elastic Net
Multilayer Perceptron	Learning Rates: 0.001, 0.01, 0.1 Activation functions: Identity, Logistic, Tanh, ReLu

Table 1. **List of experiments** performed by tuning Hyperparameters

3.2 Output Analysis and Comparison of Architectures

The averages of all the performance metric parameters were taken in all the variable hyperparameters like activation function, learning rate, and penalties as given in Table 2 for each experiment. The baseline and stochastic perceptron experiments consistently yielded identical results, as the methodologies employed were nearly similar, except that the stochastic perceptron incorporated shuffling and updated weights after processing a single training sample. In contrast, the baseline perceptron processed the entire dataset at once.

The accuracies were predominantly around 68 percent, but the losses were approximately 6.5, indicating that there are substantial losses. Ideally, losses should be closer to 0. Significant losses suggest underfitting, meaning the model performs inadequately in identifying patterns [19, 20]. The sensitivities approached 60

percent, specificities were approximately 76 percent, and the rates of false positives, false negatives, and discoveries were sufficient to render these two perceptron variant models empirical.

The weighted perceptron outperformed the baseline and stochastic models regarding test and training losses, which were approximately 5.3, indicating relatively reduced underfitting compared to the latter two models [19, 20]. The accuracies, nearing 66 percent, were not significantly distant, as class weights were adjusted to address class imbalances. Although the specificities were inferior to those of baseline and stochastic models, it was clear that the sensitivity percent had greatly increased. Nonetheless, the error classification rates, including the false positive rate (FPR) and false discovery rate (FDR), increased while the false negative rate (FNR) decreased, hence enhancing the model's reliability compared to the preceding two models concerning losses and false negative rates.

The perceptron model, utilizing the synthetic minority oversampling method (SMOTE) to ensure a balanced dataset for classifying minority (diabetic) and majority (non-diabetic) classes, demonstrated comparable accuracy to the preceding three models, achieving approximately 68 percent accuracy. The training loss was approximately 7.1, while the test loss was 6.2, suggesting that despite regularisation, there is greater underfitting compared to the preceding three models. The Perceptron utilizing SMOTE surpasses the preceding three models in sensitivity, achieving a rate of 73.18 percent. This indicates that the algorithm has enhanced its ability to predict genuine diabetes patients throughout the entire dataset. The sensitivity was lower than in the previously conducted studies, indicating that this model is underperforming in identifying non-diabetic patients within the entire dataset. The false positive rate was elevated, whereas the false negative and discovery rates were diminished. This model is applicable when the objective is to achieve high sensitivity and low false negative and discovery rates.

The Multilayer Perceptron had the highest accuracy among all evaluated models, achieving about 83 percent for training data and 72.67 percent for testing data, while also recording the lowest losses, below 1, for both training and testing phases. The sensitivity was nearly equivalent to that of the weighted perceptron, while the specificity was the greatest among all the models tested.

Model	Train Accuracy	Test Accuracy	Train Loss	Test Loss	Sensitivity (Recall)	Specificity	FPR	FNR	FDR
Baseline Perceptron	67.18%	69.70%	6.787131583	6.259466417	58.03%	76.18%	23.82%	41.97%	42.25%
Stochastic Perceptron	67.18%	69.70%	6.787132	6.259466	58.03%	76.18%	23.82%	41.97%	42.25%
Weighted Perceptron	66.15%	66.77%	5.578528	5.151272	67.58%	66.33%	33.67%	32.42%	46.86%
Perceptron with SMOTE	67.65%	68.54%	7.137558	6.262889	73.18%	63.80%	36.20%	26.82%	32.32%
Multilayer Perceptron	83.22%	72.67%	0.359184	0.661521	65.15%	76.85%	23.15%	34.85%	38.31%

Table 2. **Averages of Performance Metrics of the Experiments**

The false positive rate has been comparatively the lowest, while the false negative and discovery rates ranged around from 34 to 39 percent. This model excels in critical performance metrics, such as accuracy and loss, attributable to the implementation of hidden layers and activation functions that mitigate overfitting, hence enhancing its efficacy for optimal regression [20].

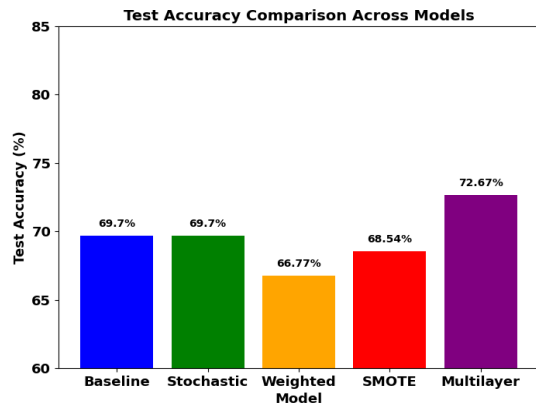


Figure 4. A Sample Bar Graph Of Comparison of Test Accuracies, for all the experiments Performed

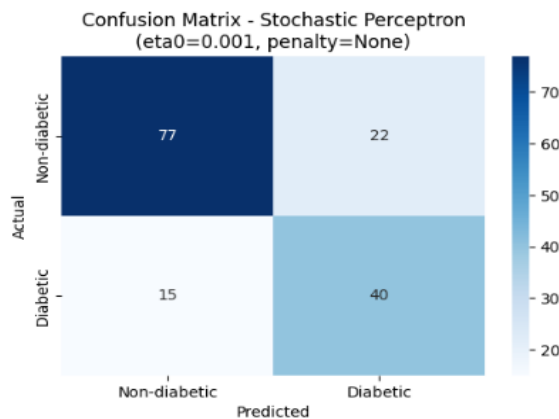


Figure 5. A sample Confusion Matrix heatmap, from which the performance metrics were calculated.

Thus, both the Baseline and the Stochastic Perceptron had a higher percentage of specificity; consequently, these are more efficient in picking out cases of diabetes. The Weighted Perceptron proved to show improvements in sensitivity, hence improving the sensitivity challenge of correctly identifying diabetic patients. The Perceptron with SMOTE enhanced sensitivity and reduced false negatives and rates of discovery, which is important for minimizing the incorrect classification of diabetic cases. Among the competitors, the best results of classification for the most accurate dataset, with the least losses and having the least false positive rate, were given by the Multilayer Perceptron. The study was further extended by illustrating all performance measures in the bar chart presented in Figure 4, while the confusion matrix facilitated the evaluation of each model's performance by comparing actual values with anticipated values, as

shown in Figure 5.

4. Code Link

The codes can be viewed at <https://github.com/a1906525/Deep-Learning-Fundamentals.git>.

5. Conclusion

To conclude, the various forms of Perceptron-based models, which have been outlined in the paper, will show strengths and weaknesses in the prediction of diabetes using the PIMA Indian dataset. As mentioned, SMOTE-applied Perceptron goes the furthest for sensitivity and few false negatives, hence most suitable for vital medical diagnosis dependent upon the correct identification of diabetic patients. Merits were present in every model here. The multi-layer perceptron showed promise in handling complex patterns in the data as it attained higher accuracy with lesser loss.

Tuning parameters, such as learning rates, penalties, and activation functions studies proved very helpful in enhancing the performance of this model. Scaling such investigations, further studies can be integrated with advanced neural network architecture or hybrid models, optimization of hyperparameters, and applications on larger and more diverse medical datasets. All these models can be scaled up further and made more robust by the incorporation of real-time data with improvement in methods of data pre-processing. All these evaluations have demarcated a potential ground for further refinement and development in future research that may lay a very sound basis for the advancement of machine learning applications in the medical domain, particularly for disease prediction.

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